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SkyeAv

Skye Lane Goetz

Applied AI/ML Engineer | Biomedical Data Scientist

Skills

Technical Skills

- **Languages:** Python, Elixir, SQL, Shell Scripting, LaTeX, Julia, Nim, Clojure
- **Data Science:** Polars, Spark, Numpy, Matplotlib, Sqlite, Postgres, DuckDB
- **AI/ML:** PyTorch, LightGBM, Scikit-learn, LangChain, MCP, Snorkel, Transformers, PEFT, ChromaDB
- **Knowledge Graphs:** Graph Databases, Ontology, KGX, Centrality, Graph Neural Networks
- **NLP:** spaCy, NLTK, Embeddings, Text Classification, TF-IDF
- **DevOps:** Nix, Docker, Setuptools, Mix, Pytest, FastAPI, Kubernetes

Domain Knowledge

- **Biology:** Heterogenous Multiomic Data, Shotgun Metagenomics, Polygenic Risk Scores, ELISA
- **Chemistry:** DFT, Mass Spectrometry, ^{1}H NMR, FTIR, Organic Synthesis, Equillibreum
- **Data Science:** PCA, KernelPCA, UMAP, Pearson Correlations, Logistic Regression, Deep Learning, ANOVA
- **Business:** SPIN Selling, Door-To-Door Sales, Product Marketing, Agribusiness, Microeconomics, Macroeconomics

Additional Competencies

- **Scientific Communication:** Lab Meetings, Conference Presentations, Journal Writing, Funding Proposals
- **Leadership and Collaboration:** Mentoring, Collaborative Learning, Team System Design, Consulting, Intermediate Spanish
- **Hobbies:** Cooking, Skiing, Geometry Dash, Gardening, Travel, Hiking

Work Experience

Institute for Systems Biology (ISB) — **Seattle, WA**
Research Associate (Data Science) [Hood Lab] — *Nov 2024 - Present*

- Published MicrobiomeKG Paper with 115,000 Edges from 40 Papers
  - Developed an Edge Scoring ResNet to Provide a Unified Comparator Across Heterogeneous Data, Weighted Enabling Centrality Analyses
- Created the MultiomicsKG with 800,000 Edges (with 52 Papers) Connecting Multiomic Factors with Host Health (also for the NCATS Biochemical Data Translator)
  - Presented Too Multiple Subject-Matter Experts to Drive Development Before Scaling to Millions of Edges
  - Worked with a Clinician (MD) to Verify Content and Suggest Improvements
- Improved and Maintained a Declarative Knowledge Extractor for Our Team
  - Reduced Knowledge Graph Generation Time from 1 Week to 8 Minutes
  - Improved Ontology Match Rates by 50%
  - Developed an AI Agent with LangChain that Drafts Configuration with Minimal Natural Language Inputs
- Co-Mentored a Clinical Trials Machine Learning Intern
  - Scaled Her Project (Classifying Untrustworthy Reporting in the Clinical Trials Database) After She Left
  - Expanded Clinical Trial Data With Embeddings (For High Cardinality Text Data)
  - Increased AUC from 0.63 to 0.90 with Expanded Data, Weak Supervision, and a New Model Architecture
  - Presented Generalized Work to Subject-Matter Experts to Drive Development of a Sucess Classifier for Early Phase Clinical Trials

Zigler-Haynes Lab (CalPoly) — **San Luis Obispo, CA**
Student Researcher (Computational Subgroup) — *Sep 2024 - Nov 2024*

- Leveraged TD-DFT Calculations to Map the Stability of Various Thiocarboxylate Derivatives
- Developed a Declarative ORCA Wrapper (in Julia) to Increase Our Labs Accessibility to DFT Calculations
  - The Wrapper was 3 Times Faster at Parsing Output into a Report than Previous Lab Scripts

Institute for Systems Biology (ISB) — **Seattle, WA**
Data Science Intern [Hood Lab] — *Aug 2024 - Sep 2024*

- Developed (Tablassert) a Declarative Framework to Build a KGX Compliant Knowledge Graph from Tabular Biomedical Data in the Supplementary Materials of Academic Papers (Mostly PubMed)
  - Improved Scalability By Eliminating the Need to Write Scripts for Every Tabular Material
- Co-Authored a Preprint on MicrobiomeKG, Prototyping PubMed Table Mining Methods

Institute for Systems Biology (ISB) **Seattle, WA**
Summer Intern/Trainee (NSF REU) [Hood-Price Lab] *Jun 2024 - Aug 2024*

- Built a 40,000 Edge KGK Compliant Knowledge Graph Linking the Microbiome and Host Health for the NCATS Biomedical Data Translator (MicrobiomeKG)
  - Data Mined Excel Spreadsheets from Supplementary Materials of Academic Papers to Build the Graph
- Presented across Institutional and National Settings to Collect Feedback from Subject-Matter Experts to Drive Development
- Collaborated in a +200 Member Consortium

CalPoly Water-Energy Sustainability Training Team (WESTT) **San Luis Obispo, CA**
Student Researcher (PHOS and CRIAS projects) *Jan 2024 - Jun 2024*

- Scaled and Maintained Algal Cultures for Wastewater Treatment and Biofertilizer in Lab and Field Settings
- Monitored Culture Growth and Contamination through Microscopy, OD Measurements, and Mass Assays

Compass Group USA **San Luis Obispo, CA**
Food Service Worker *Jan 2024 - Jun 2024*

- Communicated with Team Members using Intermediate Spanish
- Occasionally Repaired Dishwashing and Kitchen Equipment

General Polymer Solutions (GPS Polymers) LLC **Upland, CA**
Customer Fulfillment and Logistics *Aug 2021 - Sep 2023*

- Streamlined Inventory Management with Improved Excel Formulation
- Coordinated Pickups with UPS, FedEx, and Other Couriers

US Soccer Federation **Upland, CA**
US Soccer Grassroots Referee (Grade 8) *Jun 2021 - Sep 2023*

- Worked with Associations and as a Private Contractor to Secure Work
- Dealt with Difficult Parents During High Stakes Games

Self Employed **Upland, CA**
Math Tutor *Jun 2023 - Sep 2023*

- Worked with Clients (Grades 3-5) to Improve Subject Performance
  - Turned Two C Students into A Students

Kessen Star Spinners **Upland, CA**
Founder, President, and CEO *Dec 2015 - Sep 2016*

- Founded and Ran a Fidget Spinner Company with 4 Employees
- Used FreeCAD to Customized Fidget Spinner Designs to Customer Specifications and for our 3D Printers
- Coordinated with Various Organizations to Sell Our Product
  - Made Over $2,000.00 in Revenue in One Night at a Local Little League Opening Day

Publications

Education

California State Polytechnic University (Cal Poly) **San Luis Obispo, CA**
BS Biochemistry *Feb 2024 - Present*

- GPA: 3.626/4.000
- Graduation Date: Spring Quarter 2026
- Consulted on Machine Learning Senior Projects for Computer Engineering Students
  - Improved AUC by 0.15
- Developed and Trialed Computational and Machine Learning Techniques for Biochemical Assays in Lab
  - Improved R^2 Coefficients by 0.08 on Average
- Trialed Modifications to Organic Syntheses in Lab
  - Speed Up Cited Procedure by 30 Minutes
- Consulted on Multinomial Classification Efforts in the Bioinformatics Lab

California State Polytechnic University, San Luis Obispo **San Luis Obispo, CA**
BS Agribusiness *Sep 2023 - Feb 2024*

- Completed 5 Months of Coursework Before Transitioning to Biochemistry
- Wrote a Comprehensive Report on the Economics of the California Almond Industry for the 2022 Growing Season

Upland High School **Upland, CA**
High School Diploma *Aug 2019 - Jun 2023*

- GPA: 4.000/4.000
- High School Soccer 4 Years
  - Team Captain 3 Years
  - Varsity Coaches Award
  - Varsity Most Improved Player Award
  - 2021-2022 Golden Boot
- Designed (with CAD) and 3D Printed Ceramic Stamps for the Ceramics Studio
- Wrote AP Physics Notes Programs for Class TI84s

Talks and Presentations

Conference and External

- **This Time on MultiomicsKG: Edge Scoring!** – NCATS Biomedical Data Translator Fall Relay (Bethesda, MD) Sep 2025
- **Gut Microbiome Signatures of Vegan, Vegetarian, and Omnivore Diets across 21,561 Individuals** – ISB Micro Journal Club (Seattle, WA) Aug 2025
- **Updates on the New MultiomicsKG** – NCATS Biomedical Data Translator Summer Relay (Bethesda, MD) Jun 2025
- **MicrobiomeKG: Extracting Relationships between the Microbiome and Host Health from Published Supplementary Materials** – Cal Poly Bailey College Research Symposium (San Luis Obispo, CA) May 2025
- **MultiomicsKG: Extracting Knowledge Assertions from Tabular Data in Supplementary Materials** – NCATS Biomedical Data Translator Spring Relay (Bethesda, MD) Feb 2025
- **MicrobiomeKG (v0.1): A Microbiome-Centric Knowledge Graph Facilitating Human Reasoning Across the Biomedical Domain** – ISB Summer Research Poster Session (Seattle, WA) Jun 2024

Group and Lab

- **This Time on MultiomicsKG: Edge Scores and Agentic Configurators** – ISB Hood Lab Meeting (Seattle, WA) Aug 2025
- **MicrobiomeKG: Connecting the Microbiome and Host Health through Knowledge Graphs** – ISB Thorsson-Shmulevich Lab Meeting (Seattle, WA) Jul 2025
- **MicrobiomeKG: Connecting the Microbiome and Host Health through Knowledge Graphs** – ISB Gibbons Lab Meeting (Seattle, WA) Jul 2025
- **Tablassert(3.0): Transforming Tabular Data Into Knowledge Graphs** – ISB Hood Lab Meeting (Seattle, WA) Feb 2025
- **MicrobiomeKG: Story of My Internship at ISB** – ISB Hood-Price Lab Meeting (Seattle, WA) Aug 2024

Projects

Tablassert — **NCATS Translator (at ISB)**
Table Mining Declarative Knowledge Graph Builder

MicrobiomeKG — **NCATS Translator (at ISB)**
A Knowledge Graph Connecting the Microbiome and Host Health

Table Config Agent — **NCATS Translator (at ISB)**
Agentic Natural Language Configuration for Tablassert-Based Knowledge Graphs

Clinical Trials Classifier — **NCATS Translator (at ISB)**
Binomical Classification for the AACT Clinical Trials Database

MultiomicsKG — **NCATS Translator (at ISB)**
A Superset of MicrobiomeKG Connecting Multiomic Factors with Host Health